

Tableau(Assignment)

Tableau Practical — Step-by-Step Documentation

Aim

To perform different data visualization operations using Tableau on Adult and Iris datasets.

Software Used

- Tableau Desktop / Tableau Public
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Datasets Used

1. Iris Dataset
 2. Adult Dataset
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STEP 1 — Install Tableau

1. Download Tableau Desktop/Public.
 2. Install the software.
 3. Open Tableau.
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STEP 2 — Load Iris Dataset

1. Open Tableau.
2. Click:

Text File

3. Select:

Iris.csv

4. Dataset gets loaded.

5. Click:

Sheet 1

(a) 1D (Linear) Data Visualization

Type Used

Histogram

Steps

1. Create new sheet.

2. Drag:

Sepal Length Cm

to:

Columns

3. Click:

Show Me

4. Select:

Histogram

5. Histogram chart is generated.

Result

1D Linear Visualization created successfully using Histogram.

(b) 2D (Planar) Data Visualization

Type Used

Scatter Plot

Steps

1. Create new sheet.
2. Drag:

Sepal Length Cm

to:

Columns

3. Drag:

Petal Length Cm

to:

Rows

4. Drag:

Species

to:

Color

5. Drag:

Id

to:

Detail

6. Scatter plot is generated.

Result

2D Planar Visualization created successfully using Scatter Plot.

(c) 3D (Volumetric) Data Visualization

Type Used

Packed Bubble Chart

Steps

1. Create new sheet.

2. Click:

Show Me

3. Select:

Packed Bubble Chart

4. Drag:

Petal Length Cm

to visualization area.

5. Drag:

Sepal Width Cm

to:

Size

6. Drag:

Species

to:

Color

7. Drag:

Id

to:

Detail

8. Packed bubble chart is generated.

Result

3D Volumetric Visualization created successfully using Packed Bubble Chart.

STEP 3 — Load Adult Dataset

1. Go to:

Data → New Data Source

2. Click:

Text File

3. Select:

adult1.csv

4. Dataset gets loaded.

(d) Temporal Data Visualization

Type Used

Line Chart

Steps

1. Create new sheet.
2. Go to:

Analysis → Create Calculated Field

3. Enter name:

Date

4. Enter formula:

TODAY()

5. Click:

OK

6. Drag:

Date

to:

Columns

7. Right click Date and select:

Exact Date

8. Again right click Date and select:

Continuous

9. Drag:

Age

to:

Rows

10. Change:

SUM(Age)

to:

AVG(Age)

11. Click:

Show Me

12. Select:

Line Chart

Result

Temporal Data Visualization created successfully using Line Chart.

(e) Multidimensional Data Visualization

Type Used

Multi-attribute Scatter Plot

Steps

1. Switch to Iris dataset.
2. Create new sheet.
3. Drag:

Sepal Length Cm

to Columns.

4. Drag:

Petal Length Cm

to Rows.

5. In Marks card:

- Drag:

Species

to Color.

- Drag:

Sepal Width Cm

to Size.

6. Change Marks type from:

Automatic

to:

Shape

7. Drag:

Petal Width Cm

to Shape.

8. Drag:

Id

to Detail.

Result

Multidimensional Visualization created successfully.

(f) Tree / Hierarchical Data Visualization

Type Used

Treemap

Steps

1. Switch to Adult dataset.
2. Create new sheet.
3. Click:

Show Me

4. Select:

Treemap

5. Drag:

Education

to:

Text

6. Drag:

Occupation

to:

Color

7. Drag:

Age

to:

Size

Result

Tree / Hierarchical Visualization created successfully using Treemap.

(g) Network Data Visualization

Type Used

Highlight Table

Steps

1. Create new sheet.
2. Drag:

Occupation

to:

Rows

3. Drag:

Education

to:

Columns

4. Click:

Show Me

5. Select:

Highlight Table

Result

Network Data Visualization created successfully using Highlight Table.

Conclusion

Different types of visualizations such as:

- 1D
- 2D
- 3D
- Temporal
- Multidimensional
- Hierarchical

- Network

were successfully created using Tableau on Iris and Adult datasets.